Aarnav Choudhary

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Technical Skills

Languages: Python, C++, JavaScript, SQL, HTML/CSS

ML/AI: PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas, LangChain, LangGraph

Web/Infrastructure: FastAPI, PostgreSQL, Docker, Kubernetes, Helm, Kafka, AWS, GCP, Databricks, Git, Linux

Education

University of California, Los Angeles **Expected: May 2028**

Bachelor of Science, Computer Science and Mathematics

- **GPA: 3.93/4.00** (Dean's honor list, 4 Quarters)
- **Extracurriculars:** William F. Sharpe Fellow, Bruin AI, Bruin Quant Traders, Founder: Olympic Robotics

Work Experience

BigML Lab @ UCLA | *Undergraduate Researcher* Los Angeles, CA | **Aug 2026-Present**

- Prof. Baharan Mirzasoleiman's lab. Efficient prompt selection for RL based post-training of LLMs.

EXL | *Data Management Intern (EXLData.ai)* New York, NY | **Summer 2026**

- Built AI agents to analyze and chunk internal repositories; developed a RAG-Fusion and KB auto-update CI/CD pipeline with LangChain and LangGraph; used by 40 employees and 100 clients; leading to 20% time saved.
- Resolved production failures across Databricks, Kubernetes, Helm, and Kafka; introduced pre-push secret scanning, Kubernetes admission controls, and CI security gates, reducing flagged security findings by 85%.

Amazon Web Services (AWS) | *AI Software Engineering Intern (contractual)* Los Angeles, CA | **Spring 2026**

- Collaborated to build production-grade RAG agent(website) using AWS AgentCore, to serve ~25k UCLA members.
- Implemented hybrid search (BM25 + semantic), Reciprocal Rank Fusion, context-injection, and user sessions; reduced latency from 38s to 5.5s via routing decisions, model selection, data caching and cron jobs for cold starts.

Diro – Internet Original Documents | *Software Engineering Intern (AI)* New York, NY | **Summer 2025**

- Worked with a team of 5 to build a production-grade AI usage tracking platform using LiteLLM, Langfuse, PostgreSQL and GCP, providing full-stack observability for company-wide AI usage, reducing costs by 28%.

Research Experience

Beyond Prompt Selection: Learning to Allocate Rollouts for Efficient GRPO Training: Under review at **ICLR 2027**

- ADDAROLL: novel RL-based LLM post-training method to maximize non-degenerate prompt selection for GRPO.

Predicting the Next State Is Not Enough: JEPA Representations for Lean Theorem Proving- **EMNLP 2026**, Math NLP

- Applied JEPA (self-supervised learning) to Lean 4 proof picker, outperforming baseline by 5 percentage points.

Asymmetric Collapse in Model Merging: When Refusal Overwrites Recognition- **COLM 2026**, AIW and Adv-ML

- Introduced novel LLM model merging technique for successful safety preserving model merging.

Projects

Akatosh AI. [Website] **Dec 2025–May 2026**

Deep Learning, REST APIs, Full-Stack Development, Vercel

- Designed and trained a deep neural network to score leads; owned end-to-end architecture, APIs, integration expansion, and Vercel deployment with focus on performance, reliability, and scalability. 50+ users from Adobe.

Cactus Compute (YC S25), Open Source Contributor. [GitHub] **Mar 2026–Present**

C++, OpenMP, quantization, KV-cache quantization, tensor graph

- Evaluated TurboQuant and KV-cache quantization strategies helping achieve a 7x memory reduction with minimal quality loss; contributed fixes and implementation proposals to the C++ inference engine.

Clubapply.ai. [GitHub] **Oct 2025**

AWS (EC2, ECS, Lambda), Python, Docker, FastAPI, S3, IAM, RAG

- Agentic platform with 8 orchestration agents for college club applications. Built multimodal mock interviewer. Winner, best use of AWS technology at AWS x BruinAI hackathon 2025.